

SCOT TOPAZ

Community Operations & Management Platform Overview

Prepared for the SCOT Team • Sports & Cultural Organizers of Topaz

The Vision: Community-First Operations

Moving Beyond Spreadsheets: Replacing fragmented WhatsApp chats, paper receipts, and spreadsheets with a single, synchronized platform.

Playful, Colorful Identity: Built with a warm, colorful, and highly approachable UI/UX. The theme represents community spirit, leaving dark tables behind.

Self-Service Resident Hub: Empowers society members to log in, track dues, verify eligibility, sign up for events, and see tournament scores.

Organizers Control Console: Provides tools for the SCOT Admin, Core Team, Wing Commanders, and Event Champions to manage society activities seamlessly.

App Core Features Directory

RESIDENT HUB

Dues & Maintenance Tracker: Instant display of flat-specific dues (Paid vs Pending) for the active season.

Event Registrations Gate: Sign up for sports & cultural events (gated by flat eligibility check).

Confirmations & Receipts: Securely download contribution statements.

ORGANIZERS CONSOLE

Admin Bootstrap: Admin manages team onboarding (Core Team, Wing Commanders, Wing Captains).

Contributions Override: Log cash or online flat payments securely.

Expense Approval Board: Approve vendor quotes and bills based on budget limits.

Tournament Scoreboard: Log match results, generate schedules, and update lead stands.

Bootstrap & Onboarding Lifecycle

Phase 1: Admin Onboard Organizers: SCOT Admin logs in to register committee members (Core Team, Wing Commanders, Wing Captains, Event Champions) and assigns their wing boundaries.

Phase 2: Organizer Onboard Residents: Wing Commanders and Wing Captains onboard residents (Home Chiefs, Home Members) in their specific wings and assign them to physical flats.

Supabase Database Onboarding Trigger: Signups use phone-only OTP auth. When a user first enters the app, their profile is automatically generated and linked to their flat from the admin metadata claims.

Financial Operations & Ledger Guardrails

Phase 3: Record Flat Contribution: Wing Commanders log flat contributions. Once recorded, the flat is marked as 'PAID', and residents of that flat are instantly unlocked to register for events.

Phase 4: Expense Approvals Board: Core Team members upload vendor invoices or quotations. The app routes approval requests dynamically based on financial thresholds:

Low-tier Expense (< ₹5,000): Automatically approved by the system.

Mid-tier Expense (₹5,000 to ₹25,000): Requires a review and signature from one Core Team member.

High-tier Expense (> ₹25,000): Requires review and signatures from two unique Core Team members.

Competition & Tournament Scoring Engine

Automated Match Scheduler: Programmatically generates Round Robin fixtures using the Circle Method, and handles Knockout brackets with power-of-two byes.

Dynamic Scoring Configuration: Win, Draw, Loss points, and tie-breaker criteria (e.g. head-to-head, participation counts) are configurable per competition.

Transaction-Safe Recording: Event Champions submit scores directly in the app. The system checks walkovers, splits points for ties, and updates standings atomically.

Participation Caps (Scenario B): Imposes point limits per Wing to reward diverse participation, preventing a single wing from dominating based on team size alone.

Non-Technical Safety & Security Guards

Row-Level Security (RLS) Isolation: Residents can only view flat payments and profiles inside their own flat. Wing Commanders and Captains can only view or manage residents within their assigned wing. No one can snoop on other wings.

Financial Auditing Trigger: Every manual contribution update or payment override is automatically recorded in an immutable ledger (`finance.audit_log`) showing exactly who modified the state, when, and the amount.

Authorized Database Access: Critical procedures (like `record_payment` or `approve_expense`) run internal role checks. A user attempting to bypass the front-end code is blocked by the database layer.

Outstanding Targets & Future Roadmap

Real-Time Push Notifications: Broadcasting announcements, payment reminders, match timings, and live score updates to residents.

Google Drive Media Gallery: Uploading and streaming high-resolution tournament photos and cultural event videos directly from a shared society Google Drive space.

Secure Voting Polls: Allows coordinators to publish polls for society resolutions, restricting voting to one vote per flat (Home Chief claim) to ensure democratic participation.

OCR-based Receipt Processing: Enabling residents to upload bank payment screenshots and auto-extracting transaction reference IDs, dates, and amounts to prevent manual entry errors.